

YOUNGHOON JUNG, 정영훈

INFORMATION

- Ph.D. of Mathematics
- Principal research engineer at Mobilint, Framework Team
- AI Compiler — IR Design & Graph Optimization for NPU Acceleration
- Seoul, South Korea
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SUMMARY

AI compiler engineer specializing in making deep learning models executable on constrained NPU hardware.

Own the compiler layer bridging model frameworks and NPU hardware, designing IR and graph transformation systems that enable execution of models previously infeasible due to static execution constraints and a restricted operator set.

Proven impact in deploying LLM, VLM, and vision models to production NPU systems with measurable performance gains.

Cross-disciplinary background in mechanical engineering, mathematical sciences, and software systems.

TECHNICAL SKILLS

Languages	Python, C++, MATLAB, Java, Scala, Julia
AI / Systems	NPU Compiler, Graph Optimization, Execution Optimization, Edge AI
Frameworks	PyTorch, TensorFlow, TFLite, ONNX
Optimization	Operator Fusion, Graph Rewriting, Scheduling, Dataflow Optimization
Mathematics	PDE, Inverse Problems, Asymptotic Analysis, Scientific Computing

EXPERIENCE

Mobilint

Jan 2024 – Present

Principal Research Engineer – Framework Team

- Own the compiler layer bridging model frameworks and NPU hardware, enabling execution of models previously infeasible under static execution constraints and limited operator support
- Led design and implementation of IR and graph transformation pipeline (rewriting, operator decomposition, fusion) to support LLM and vision models under hardware constraints
- Enabled execution of LLM/VLM models on static-graph NPU by bypassing the Torch→ONNX pipeline and directly lowering models into a custom IR and custom graph transformation strategies
- Expanded model support by converting non-native operators into executable subgraphs under hardware constraints
- Improved 6—17% FPS improvement across vision models through graph-level dataflow optimization

- Delivered end-to-end model execution on resource-constrained NPU in collaboration with quantization and backend compiler teams

Mobilint

Jan 2022 – Dec 2023

Principal Research Engineer – Compiler Team

- Re-architected a monolithic script-level backend compiler into a structured compiler pipeline for edge AI accelerators
- Built NPU module-level simulator for performance validation using SystemC

Samsung SDS

Jan 2021 – Dec 2021

Senior Engineer – Platform Advanced Research Lab

- Built a GPU workload trace modeling pipeline to generate realistic large-scale cluster workloads
- Led development of a GPU cluster scheduling simulator driven by workload traces to evaluate scheduling strategies
- Led design and implementation of a reinforcement learning-based job scheduler on top of the simulator to optimize resource allocation and throughput

Samsung SDS

Mar 2019 – Dec 2020

Senior Engineer – Analytics Platform Lab

- Developed distributed machine learning components on Spark for the Brightics AI platform
- Built a high-performance SQL execution engine on Pandas with Apache Calcite and a NumPy-based backend, improving memory efficiency and execution performance
- Contributed to AutoML pipeline development

EDUCATION

SEP 2014 - FEB 2019	Ph.D. in MATHEMATICAL SCIENCES, KAIST , Korea Advisor: Mikyoung Lim
SEP 2012 - AUG 2014	M.S. in MATHEMATICAL SCIENCES, KAIST , Korea Advisor: Mikyoung Lim
FEB 2008 - AUG 2012	B.E. in MECHANICAL ENGINEERING, KAIST , Korea Double Major in MATHEMATICAL SCIENCES
MAR 2005 - FEB 2008	Korea Science Academy , Korea

EXPERIENCE

Teaching Assistant

Sep. 2012 - Dec. 2018

- Undergraduate courses - Analysis I, Analysis II, Fourier Analysis, Introduction to Differential Geometry, Introduction to Linear Algebra, Calculus I, Calculus II.
- Graduate courses - Real Analysis, Complex Analysis.

Coursera staff, TA

2017

- Introduction to Ordinary Differential Equations (Prof. Kwon.)

KAIST OLEV Internship

Summer 2011

- Designed a mechanical structure and conducted a thermal analysis of battery module of an online electric vehicle.

PUBLICATIONS AND PREPRINTS

- [1] Spectral analysis of the Neumann–Poincaré operator on the crescent-shaped domain and touching disks and analysis of plasmon resonance, **YH Jung**, M Lim. (2023) **Nonlinear Analysis: Real World Applications** 74, 103951
- [2] Series expansions of the layer potential operators using the Faber polynomials and their applications to the transmission problem, **Y Jung**, M Lim. (2021) **SIAM Journal on Mathematical Analysis** 53 (2), 1630-1669.
- [3] A decay estimate for the eigenvalues of the Neumann-Poincaré operator using the Grunsky coefficients, **YH Jung**, M Lim. (2020) **Proceedings of the American Mathematical Society** 148 (2), 591-600
- [4] Numerical solution to the interface problem in a general domain using Moser’s deformation method, E Hong, E Lee, **Y Jung**, M Lim. (2020) **Journal of Applied Mathematics and Computing** 65 (1), 379-401.
- [5] A joint sparse recovery framework for accurate reconstruction of inclusions in elastic media. Yoo, J., **Jung, Y.**, Lim, M., Ye, J. C., and Wahab, A. (2017). **SIAM Journal on Imaging Sciences**, 10(3), 1104-1138.

PRESENTATIONS

- [1] Series expansion of single layer potential and Neumann-Poincare operator, contributed talk, **KSIAM 2018 Annual Meeting**, Jeju, Korea.
- [2] Series representation of layer potential operators for the transmission problem, contributed talk, **ICIP 2018 Singapore**, Singapore.